

Supplementary Material

PhysioRisk: an interpretable survival-modeling framework for separating demographic and physiological mortality risk

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Numerical values are transcribed from the cited canonical artifacts and rounded only for display; source artifacts retain full precision.

Supplementary Table S1. Paired C-index contrasts in the temporal-validation cohort

Contrast	Cohort	ΔC (A – B)	95% CI low	95% CI high	Bootstrap proportion $\Delta C > 0$	Replicates	Seed
CalibratedTotalRisk – DemoRisk	temporal_test	0.01628	0.01109	0.02199	1.0	2000	20260821
CalibratedTotalRisk – PhenoAge NCP	temporal_test	0.00205	–0.00326	0.00748	0.776	2000	20260821
PhysioRisk – PhenoAge NCP acceleration	temporal_test	0.01625	–0.00049	0.03285	0.97	2000	20260821

Note: Paired percentile intervals were obtained from 2,000 participant bootstrap replicates in the temporal-validation cohort. Contrast estimates and interval limits are shown to the manuscript's five-decimal reporting precision; the canonical source artifact retains full precision.

Provenance: tables/cindex_bootstrap_contrasts.csv.

Supplementary Table S2. Proportional-hazards diagnostics

Model	Cohort	N	Deaths	Time transform	Test statistic	P value	Material violation	Flag rule	Cox warning	Scaled-residual region summary
DemoRisk	temporal_test	6,739	416	rank	2.925	0.08720586	False	p<0.01 (prespecified descriptive flag; no multiplicity adjustment)	none	early_0_36m: mean=-0.087; middle_37_72m: mean=0.044; late_73m_plus: mean=0.099
PhysioRisk	temporal_test	6,739	416	rank	2.467	0.11627106	False	p<0.01 (prespecified descriptive flag; no multiplicity adjustment)	none	early_0_36m: mean=-0.035; middle_37_72m: mean=0.013; late_73m_plus: mean=0.052
CaibratedTotalRisk	temporal_test	6,739	416	rank	16.539	0.00004765	True	p<0.01 (prespecified descriptive flag; no multiplicity adjustment)	none	early_0_36m: mean=-0.177; middle_37_72m: mean=0.108; late_73m_plus: mean=0.161
HematologicStress	temporal_test	6,739	416	rank	3.852	0.04967436	False	p<0.01 (prespecified descriptive flag; no multiplicity adjustment)	none	
RenalHepaticTurnover	temporal_test	6,739	416	rank	4.304	0.03802453	False	p<0.01 (prespecified descriptive flag; no multiplicity adjustment)	none	

Note: The prespecified descriptive flag was $p<0.01$ without multiplicity adjustment. The regional scaled-residual summaries are reported where present in the canonical artifact.

Provenance: tables/ph_diagnostics.csv.

Supplementary Table S3. Diagnostic log-time interaction analyses

Model	N	Deaths	Score scale	Time interaction	Reference time (months)	Log HR/SD at 60m	HR/SD at 12m	HR/SD at 60m	HR/SD at 96m	Interaction coefficient	HR multiplier/e-fold time	95% CI low	95% CI high	P value	Direction
DemoRisk	6,739	416	actual temporal-cohort SD	score_z * log(time_months/60)	60	1.358	3.399	3.890	4.047	0.084	1.087	−0.055	0.223	0.23794	effect_strengthens_over_time
PhysioRisk	6,739	416	actual temporal-cohort SD	score_z * log(time_months/60)	60	0.356	1.366	1.428	1.446	0.027	1.028	−0.032	0.087	0.36401	effect_strengthens_over_time
CalibratedTotalRisk	6,739	416	actual temporal-cohort SD	score_z * log(time_months/60)	60	1.320	2.842	3.744	4.057	0.171	1.187	0.066	0.277	0.00150	effect_strengthens_over_time

Note: These analyses are diagnostic only and did not replace or refit the fixed predictors used in temporal validation.

Provenance: tables/ph_time_interaction.csv.

Supplementary Table S4. Fixed-horizon calibration of CalibratedTotalRisk

Horizon (months)	Training baseline survival	Mean predicted mortality	KM observed mortality	Observed – predicted	O/E ratio	Global Cox slope	Global slope CI low	Global slope CI high	Horizon intercept	Horizon slope	IPCW Brier score	IPCW usable N	No temporal recalibration
60	0.983687	0.046101	0.046533	0.000432	1.009	0.811	0.762	0.859	−0.101	0.941	0.03776	4,709	True
96	0.966595	0.085710	0.088256	0.002546	1.030	0.811	0.762	0.859	0.067	0.999	0.06018	1,518	True

Note: Absolute risks used the development-derived coefficients and baseline survival. Horizon-specific logistic calibration and Brier scores used temporal-test censoring weights for evaluation only; no temporal recalibration occurred.

Provenance: tables/calibration_fixed_horizons.csv.

Supplementary Table S5. Censoring-robust discrimination diagnostics

Model	Evaluation cohort	Censoring-distribution cohort	Training N for censoring	Test N	Horizon (months)	Uno C (IPCW)	Cumulative/dynamic AUC (IPCW)	Implementation
DemoRisk	temporal_test	training (1999-2010)	13,090	6,739	60	0.823	0.834	scikit-survival 0.28.0
DemoRisk	temporal_test	training (1999-2010)	13,090	6,739	96	0.827	0.853	scikit-survival 0.28.0
PhysioRisk	temporal_test	training (1999-2010)	13,090	6,739	60	0.654	0.657	scikit-survival 0.28.0
PhysioRisk	temporal_test	training (1999-2010)	13,090	6,739	96	0.654	0.677	scikit-survival 0.28.0
CalibratedTotalRisk	temporal_test	training (1999-2010)	13,090	6,739	60	0.839	0.850	scikit-survival 0.28.0
CalibratedTotalRisk	temporal_test	training (1999-2010)	13,090	6,739	96	0.843	0.870	scikit-survival 0.28.0
PhenoAge NCP	temporal_test	training (1999-2010)	13,090	6,739	60	0.838	0.850	scikit-survival 0.28.0
PhenoAge NCP	temporal_test	training (1999-2010)	13,090	6,739	96	0.841	0.875	scikit-survival 0.28.0
PhenoAge NCP acceleration	temporal_test	training (1999-2010)	13,090	6,739	60	0.641	0.655	scikit-survival 0.28.0
PhenoAge NCP acceleration	temporal_test	training (1999-2010)	13,090	6,739	96	0.638	0.698	scikit-survival 0.28.0

Note: The development cohort estimated the censoring distribution. These were diagnostics of fixed temporal-test predictions, not model-training steps.

Provenance: tables/censoring_robust_discrimination.csv.

Supplementary Table S6. Cycle-specific preselection biomarker missingness and cohort provenance

Cycle start year	Biomarker	Mortality-eligible N	Missing N	Missing (%)	Denominator population
1999	LBXSAL	5,445	826	15.170	mortality_eligible_before_complete_case_restriction
1999	LBXSCR	5,445	826	15.170	mortality_eligible_before_complete_case_restriction
1999	LBXGLU	5,445	3,168	58.182	mortality_eligible_before_complete_case_restriction
1999	LBXLYPCT	5,445	762	13.994	mortality_eligible_before_complete_case_restriction
1999	LBXMCV	5,445	732	13.444	mortality_eligible_before_complete_case_restriction
1999	LBXRDW	5,445	732	13.444	mortality_eligible_before_complete_case_restriction
1999	LBXSAPSI	5,445	826	15.170	mortality_eligible_before_complete_case_restriction
1999	LBXWBCSI	5,445	732	13.444	mortality_eligible_before_complete_case_restriction
2001	LBXSAL	5,987	778	12.995	mortality_eligible_before_complete_case_restriction
2001	LBXSCR	5,987	778	12.995	mortality_eligible_before_complete_case_restriction
2001	LBXGLU	5,987	778	12.995	mortality_eligible_before_complete_case_restriction
2001	LBXLYPCT	5,987	717	11.976	mortality_eligible_before_complete_case_restriction
2001	LBXMCV	5,987	690	11.525	mortality_eligible_before_complete_case_restriction
2001	LBXRDW	5,987	690	11.525	mortality_eligible_before_complete_case_restriction
2001	LBXSAPSI	5,987	778	12.995	mortality_eligible_before_complete_case_restriction
2001	LBXWBCSI	5,987	690	11.525	mortality_eligible_before_complete_case_restriction
2003	LBXSAL	5,610	650	11.586	mortality_eligible_before_complete_case_restriction
2003	LBXSCR	5,610	650	11.586	mortality_eligible_before_complete_case_restriction
2003	LBXGLU	5,610	650	11.586	mortality_eligible_before_complete_case_restriction
2003	LBXLYPCT	5,610	591	10.535	mortality_eligible_before_complete_case_restriction
2003	LBXMCV	5,610	561	10.000	mortality_eligible_before_complete_case_restriction
2003	LBXRDW	5,610	561	10.000	mortality_eligible_before_complete_case_restriction
2003	LBXSAPSI	5,610	650	11.586	mortality_eligible_before_complete_case_restriction
2003	LBXWBCSI	5,610	561	10.000	mortality_eligible_before_complete_case_restriction
2005	LBXSAL	5,561	586	10.538	mortality_eligible_before_complete_case_restriction
2005	LBXSCR	5,561	586	10.538	mortality_eligible_before_complete_case_restriction
2005	LBXGLU	5,561	3,137	56.411	mortality_eligible_before_complete_case_restriction
2005	LBXLYPCT	5,561	557	10.016	mortality_eligible_before_complete_case_restriction
2005	LBXMCV	5,561	535	9.621	mortality_eligible_before_complete_case_restriction
2005	LBXRDW	5,561	535	9.621	mortality_eligible_before_complete_case_restriction
2005	LBXSAPSI	5,561	586	10.538	mortality_eligible_before_complete_case_restriction
2005	LBXWBCSI	5,561	535	9.621	mortality_eligible_before_complete_case_restriction
2007	LBXSAL	6,219	655	10.532	mortality_eligible_before_complete_case_restriction
2007	LBXSCR	6,219	656	10.548	mortality_eligible_before_complete_case_restriction
2007	LBXGLU	6,219	3,468	55.765	mortality_eligible_before_complete_case_restriction
2007	LBXLYPCT	6,219	626	10.066	mortality_eligible_before_complete_case_restriction
2007	LBXMCV	6,219	612	9.841	mortality_eligible_before_complete_case_restriction
2007	LBXRDW	6,219	612	9.841	mortality_eligible_before_complete_case_restriction
2007	LBXSAPSI	6,219	657	10.564	mortality_eligible_before_complete_case_restriction
2007	LBXWBCSI	6,219	613	9.857	mortality_eligible_before_complete_case_restriction
2009	LBXSAL	6,510	561	8.618	mortality_eligible_before_complete_case_restriction
2009	LBXSCR	6,510	561	8.618	mortality_eligible_before_complete_case_restriction
2009	LBXGLU	6,510	3,581	55.008	mortality_eligible_before_complete_case_restriction
2009	LBXLYPCT	6,510	500	7.680	mortality_eligible_before_complete_case_restriction
2009	LBXMCV	6,510	491	7.542	mortality_eligible_before_complete_case_restriction
2009	LBXRDW	6,510	491	7.542	mortality_eligible_before_complete_case_restriction
2009	LBXSAPSI	6,510	561	8.618	mortality_eligible_before_complete_case_restriction
2009	LBXWBCSI	6,510	491	7.542	mortality_eligible_before_complete_case_restriction

Cycle start year	Biomarker	Mortality-eligible N	Missing N	Missing (%)	Denominator population
2011	LBXSAL	5,849	700	11.968	mortality_eligible_before_complete_case_restriction
2011	LBXSCR	5,849	700	11.968	mortality_eligible_before_complete_case_restriction
2011	LBXGLU	5,849	3,254	55.633	mortality_eligible_before_complete_case_restriction
2011	LBXLYPCT	5,849	545	9.318	mortality_eligible_before_complete_case_restriction
2011	LBXMCV	5,849	540	9.232	mortality_eligible_before_complete_case_restriction
2011	LBXRDW	5,849	540	9.232	mortality_eligible_before_complete_case_restriction
2011	LBXSAPSI	5,849	702	12.002	mortality_eligible_before_complete_case_restriction
2011	LBXWBCSI	5,849	540	9.232	mortality_eligible_before_complete_case_restriction
2013	LBXSAL	6,100	482	7.902	mortality_eligible_before_complete_case_restriction
2013	LBXSCR	6,100	482	7.902	mortality_eligible_before_complete_case_restriction
2013	LBXGLU	6,100	3,377	55.361	mortality_eligible_before_complete_case_restriction
2013	LBXLYPCT	6,100	430	7.049	mortality_eligible_before_complete_case_restriction
2013	LBXMCV	6,100	414	6.787	mortality_eligible_before_complete_case_restriction
2013	LBXRDW	6,100	414	6.787	mortality_eligible_before_complete_case_restriction
2013	LBXSAPSI	6,100	483	7.918	mortality_eligible_before_complete_case_restriction
2013	LBXWBCSI	6,100	414	6.787	mortality_eligible_before_complete_case_restriction
2015	LBXSAL	5,974	600	10.044	mortality_eligible_before_complete_case_restriction
2015	LBXSCR	5,974	601	10.060	mortality_eligible_before_complete_case_restriction
2015	LBXGLU	5,974	3,406	57.014	mortality_eligible_before_complete_case_restriction
2015	LBXLYPCT	5,974	564	9.441	mortality_eligible_before_complete_case_restriction
2015	LBXMCV	5,974	563	9.424	mortality_eligible_before_complete_case_restriction
2015	LBXRDW	5,974	563	9.424	mortality_eligible_before_complete_case_restriction
2015	LBXSAPSI	5,974	601	10.060	mortality_eligible_before_complete_case_restriction
2015	LBXWBCSI	5,974	563	9.424	mortality_eligible_before_complete_case_restriction

Note: Missingness was quantified within each cycle among mortality-eligible participants before complete-case restriction. The denominator is repeated for each biomarker within a cycle; biomarker-specific missing counts are not joint complete-case exclusions.

Provenance: tables/missingness_by_cycle.csv.

Supplementary Table S7. Mortality-eligible participants included in versus excluded from the benchmark cohort

Group	Population definition	N	Age mean	Age SD	Female (%)	Deaths N	Deaths (%)	Median follow-up (months)
benchmark_included	Complete for notebook 01 EXTENDED_16	19,829	47.009	19.050	50.643	2,936	14.807	124
excluded_before_benchmark	Missing one or more notebook 01 EXTENDED_16 fields	33,426	47.784	19.862	52.486	6,168	18.453	122

Note: Inclusion was defined by completeness for notebook 01’s EXTENDED_16 fields. The two groups partition the 53,255 mortality-eligible participants: 19,829 included and 33,426 excluded.

Provenance: tables/cohort_included_excluded_characteristics.csv.

Supplementary Table S8. Common complete-case and training-derived-imputation sensitivity analyses

Analysis	Score	Cohort	N	Deaths	C-index	Age correlation	Missing-data rule	Status
common_complete_case	PhysioRisk	temporal_test	6,739	416	0.654	0.106	identical participants; all no-CRP comparator and PhysioRisk biomarkers observed	performed
common_complete_case	PhenoAge NCP (no-CRP proxy)	temporal_test	6,739	416	0.841	0.961	identical participants; all no-CRP comparator and PhysioRisk biomarkers observed	performed
common_training_derived_imputation	PhysioRisk	temporal_test	6,739	416	0.654	0.106	no-op in retained benchmark because all required biomarkers are observed; upstream imputation cannot be reconstructed	performed as degenerate no-missing sensitivity; extension to excluded source participants not estimable without frozen residualization parameters
common_training_derived_imputation	PhenoAge NCP (no-CRP proxy)	temporal_test	6,739	416	0.841	0.961	no-op in retained benchmark because all required biomarkers are observed; upstream imputation cannot be reconstructed	performed as degenerate no-missing sensitivity; extension to excluded source participants not estimable without frozen residualization parameters

Note: All 6,739 temporal-validation participants in the retained benchmark already had the biomarkers required for both scores. Common training-derived imputation was therefore a no-op and changed no observations; extension to excluded source participants was not estimable from the frozen retained scoring path.

Provenance: tables/comparator_sensitivity.csv.

Supplementary Table S9. Historical submitted benchmark results retained for provenance only

Status	Interpretation	Cohort	Risk score	C-index	HR per 1 SD	HR CI low	HR CI high	P value	Age correlation
historical_submitted_not_current	provenance_only; PhysioRisk HR did not converge and TotalRisk_2axis is the obsolete unit-weight composite	test_benchmark	BaselineRisk	0.827	3.736	3.347	4.170	4.38e-122	0.970
historical_submitted_not_current	provenance_only; PhysioRisk HR did not converge and TotalRisk_2axis is the obsolete unit-weight composite	test_benchmark	PhysioRisk_2axis	0.654	2.772	2.656	2.893	0	0.106
historical_submitted_not_current	provenance_only; PhysioRisk HR did not converge and TotalRisk_2axis is the obsolete unit-weight composite	test_benchmark	TotalRisk_2axis	0.840	1.877	1.815	1.942	1.08e-293	0.811
historical_submitted_not_current	provenance_only; PhysioRisk HR did not converge and TotalRisk_2axis is the obsolete unit-weight composite	test_benchmark	PhenoAge_noCRP_proxy	0.841	3.976	3.637	4.347	3.4e-202	0.961
historical_submitted_not_current	provenance_only; PhysioRisk HR did not converge and TotalRisk_2axis is the obsolete unit-weight composite	test_benchmark	PhenoAgeAccel	0.638	1.432	1.360	1.508	3.7e-42	0.033
historical_submitted_not_current	provenance_only; PhysioRisk HR did not converge and TotalRisk_2axis is the obsolete unit-weight composite	train_benchmark	BaselineRisk	0.855	4.286	4.097	4.484	0	0.971
historical_submitted_not_current	provenance_only; PhysioRisk HR did not converge and TotalRisk_2axis is the obsolete unit-weight composite	train_benchmark	PhysioRisk_2axis	0.602	1.334	1.295	1.374	1.07e-81	-0.000
historical_submitted_not_current	provenance_only; PhysioRisk HR did not converge and TotalRisk_2axis is the obsolete unit-weight composite	train_benchmark	TotalRisk_2axis	0.858	2.427	2.372	2.483	0	0.790
historical_submitted_not_current	provenance_only; PhysioRisk HR did not converge and TotalRisk_2axis is the obsolete unit-weight composite	train_benchmark	PhenoAge_noCRP_proxy	0.864	3.670	3.556	3.787	0	0.958
historical_submitted_not_current	provenance_only; PhysioRisk HR did not converge and TotalRisk_2axis is the obsolete unit-weight composite	train_benchmark	PhenoAgeAccel	0.586	1.246	1.219	1.274	1.13e-85	0.000

Note: Historical provenance only. The historical PhysioRisk hazard-ratio fit did not converge, and TotalRisk_2axis is the obsolete unit-weight composite. These rows are not current manuscript results.

Provenance: manuscript/generated/table_s9_historical_submitted_results.csv (generated by scripts/build_s4_manuscript_assets.py from tables/physiorisk_vs_phenoage_aligned_comparison.csv).